CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

Case 10-2025: A 32-Year-Old Woman with Flank Pain, Fever, and Hypoxemia

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PRESENTATION OF CASE

Dr. Margot E. Manning (Medicine): A 32-year-old woman was admitted to this hospital because of flank pain, fever, and hypoxemia.

Two weeks before the current presentation, sharp, intermittent pain in the left flank developed. Two days before the current presentation, the pain worsened in intensity, became more constant, and was associated with nausea. The next day, the patient had multiple episodes of vomiting. On the morning of the current presentation, she noted a dry cough and a subjective feeling of warmth, although she had not checked her temperature at home. She presented to an urgent care clinic affiliated with her primary care physician's office for evaluation.

In the urgent care clinic, the patient reported ongoing pain in the left flank and dry cough. No increased urinary frequency, dysuria, or hematuria was present, and she had no shortness of breath. The temporal temperature was 38.1°C, the blood pressure 107/75 mm Hg, the pulse 113 beats per minute, the respiratory rate 20 breaths per minute, and the oxygen saturation 95% while she was breathing ambient air. Physical examination revealed abdominal tenderness on palpation of the left lower quadrant, but no tenderness was detected when the costovertebral angles were tapped. The white-cell count was 19,000 per microliter (reference range, 3800 to 10,800) with a neutrophilic predominance. The blood level of albumin was 2.8 g per deciliter (reference range, 3.3 to 5.5), and carbon dioxide 23 mmol per liter (reference range, 24 to 29); the anion gap was 18 mmol per liter (reference range, 10 to 20). Urinalysis showed 3+ blood (reference value, negative) and 3+ protein (reference value, negative) but was otherwise normal. A urine test for human chorionic gonadotropin was negative. Results of other laboratory tests are shown in Table 1.

The oxygen saturation decreased to 89% while the patient was in the urgent care clinic and then increased to 94% when oxygen was delivered through a nasal cannula at a rate of 3 liters per minute. She was transported by ambulance to the emergency department of this hospital.

Table 1. Laboratory Data.*

Variable	Reference Range, Adults, Urgent Care Clinic	On Initial Presentation, Urgent Care Clinic	Reference Range, Adults, This Hospital†	On Current Presentation, This Hospital
Sodium (mmol/liter)	138–146	138	135–145	138
Potassium (mmol/liter)	3.5–4.9	3.7	3.4–5.0	3.5
Chloride (mmol/liter)	98–109	101	98–108	104
Carbon dioxide (mmol/liter)	24–29	23	23–32	22
Urea nitrogen (mg/dl)	8–26	11	8–25	11
Creatinine (mg/dl)	0.6–1.3	1.1	0.60–1.50	0.93
Glucose (mg/dl)	70–105	147	70–110	127
Anion gap (mmol/liter)	10–20	18	3–17	12
Calcium (mg/dl)	—	—	8.5–10.5	8.0
Hematocrit (%)	35.0–45.0	40.7	36.0–46.0	42.7
Hemoglobin (g/dl)	11.7–15.5	13.5	12.0–16.0	14.1
White-cell count (per μl)	3800–10,800	19,000	4500–11,000	16,020
Differential count (per μl)				
Neutrophils	1500–7800	16,460	1800–7700	14,210
Lymphocytes	850–3900	1400	1000–4800	1240
Monocytes	200–950	1120	200–1200	400
Eosinophils	20–500	10	0–900	0
Basophils	0–200	10	0–300	40
Bands	—	—	0–100	130
Platelet count (per μl)	140,000–400,000	190,000	150,000–400,000	173,000
Alkaline phosphatase (U/liter)	42–141	88	30–100	104
Alanine aminotransferase (U/liter)	10–47	19	7–33	18
Aspartate aminotransferase (U/liter)	11–38	14	9–32	17
Total bilirubin (mg/dl)	0.2–1.6	0.7	0.0–1.0	0.4
Direct bilirubin (mg/dl)	—	—	0.0–0.3	<0.2
Total protein (g/dl)	6.4–8.1	6.6	6.0–8.3	7.0
Albumin (g/dl)	3.3–5.5	2.8	3.3–5.0	3.3
Lactic acid (mmol/liter)	—	—	0.5–2.0	2.2
Lactate dehydrogenase (U/liter)	—	—	110–210	278
International normalized ratio	—	—	0.9–1.1	1.2
C-reactive protein (mg/liter)	—	—	0–8.0	199.9
Erythrocyte sedimentation rate (mm/hr)	—	—	0–19	40

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for lactic acid to milligrams per deciliter, divide by 0.1110.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

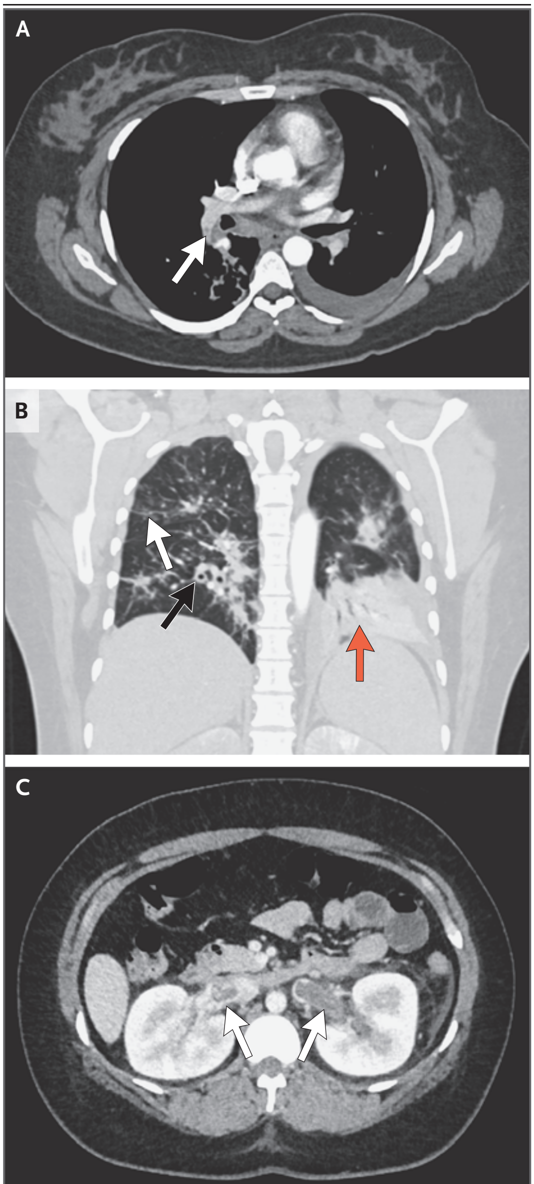

Figure 1. Contrast-Enhanced CT Images of the Chest and Abdomen.

An axial image of the chest (Panel A) shows poor opacification of the pulmonary arteries; however, a partial filling defect is present in the right lower lobar pulmonary artery (arrow). A coronal image of the chest in a lung window (Panel B) shows bronchial wall thickening (black arrow), patchy peribronchiolar ground-glass opacities (white arrow), and dependent consolidation and atelectasis in the left lower lobe (orange arrow). An axial image of the abdomen and pelvis (Panel C) shows a nearly occlusive thrombus in the left renal vein and a nonocclusive thrombus in the right renal vein (arrows).

In the emergency department, the patient reported ongoing left flank pain, recurrent nausea, cough with new sputum production, and new shortness of breath. There was no history of joint pain, joint swelling, oral ulcers, or rash. She had no relevant medical history and had not been pregnant. Her surgical history included appendectomy. The patient took no medications and had no known adverse reactions to medications. She worked as a gardener and lived with a sister in Massachusetts. She did not smoke tobacco, vape, or use illicit drugs. She drank alcohol occasionally. Both of her parents had diabetes and hypertension. One sister had had kidney disease that required hemodialysis at 17 years of age; she died at 18 years of age. The patient identified as Hispanic. There was no family history of blood clots or cancer.

On examination, the temporal temperature was 39.6°C, the blood pressure 102/58 mm Hg, the pulse 122 beats per minute, the respiratory rate 28 breaths per minute, and the oxygen saturation 96% while she was receiving oxygen through a nasal cannula at a rate of 3 liters per minute. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 30.1. The patient appeared uncomfortable. No oral ulcers, alopecia, or rash was present. The results of a joint examination were normal. On auscultation, inspiratory and expiratory course crackles were noted in both lungs. The heart sounds were tachycardic but otherwise normal. She had pain on palpation of the left lower abdomen. No edema was present.

Results of laboratory tests are shown in Table 1. Urinalysis revealed 3+ blood (reference value, negative), 3+ protein (reference value, negative), more than 100 red cells per high-power field (reference range, 0 to 2), and 10 to 20 white cells per high-power field (reference value, <10). The ratio of protein to creatinine in urine was 3.5 (reference value, <0.15), a finding that suggests a urinary protein level of approximately 3.5 g per day. The glycated hemoglobin level was 6.2% (reference range, 4.3 to 5.6). A test for severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) RNA was negative. Blood was obtained for culture, and imaging studies were performed.

Dr. Cynthia L. Czawlytko: Chest radiography revealed low lung volumes; patchy basilar and retrocardiac opacities with partial obscuration of the left hemidiaphragmatic border and a small pleural effusion were observed on the left side. Computed tomography (CT) of the chest, performed according to a pulmonary embolism protocol, was markedly limited for the detection of pulmonary emboli because of the phase of contrast material but showed a partial filling defect in the right lower lobar pulmonary artery and a suspected filling defect in a subsegmental artery in the right lower lobe (Fig. 1A). Bronchial wall thickening was present, as were multifocal peribronchiolar and dependent consolidative opacities with surrounding ground-glass opacities (Fig. 1B). Small left and trace right pleural effusions and minimal areas of interlobular septal thickening associated with areas of ground-glass opacities were observed. CT of the abdomen and pelvis, performed after the administration of intravenous contrast material, revealed a nearly occlusive thrombus in the left renal vein and a nonocclusive thrombus in the right renal vein (Fig. 1C) without thrombus or compression in the iliac veins. Early heterogeneous parenchymal enhancement of the kidneys and moderate perinephric stranding were observed. No evidence of cancer was seen in the chest, abdomen, or pelvis.

Dr. Manning: The patient received 2 liters of lactated Ringer's solution, and treatment with intravenous vancomycin and cefepime was started.

The patient was admitted to the hospital, and a diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Anushya Jeyabalan: I participated in the care of this patient, and I am aware of the final diagnosis. This previously healthy young woman presented with severe flank pain, fever, and hypoxemia. Notable laboratory findings included leukocytosis, hematuria, heavy proteinuria, and hypoalbuminemia. At the urgent care clinic, the blood level of creatinine was reported within the normal range, but without a baseline creatinine level, it is possible to miss acute kidney injury — particularly in young patients. Monitoring the blood levels of creatinine over time would be essential to determine if this patient had acute kidney injury on admission. Imaging revealed multiple venous thromboemboli, including a suspected pulmonary embolism and bilateral renal vein thromboses. Areas of interlobular septal thickening with associated ground-glass opacity were observed, a finding that could be consistent with diffuse alveolar hemorrhage. Given these findings in the context of concurrent hematuria and proteinuria, I will begin by considering possible causes of pulmonary–renal syndrome.

PULMONARY–RENAL SYNDROME

Pulmonary–renal syndrome is characterized by the combination of diffuse alveolar hemorrhage and glomerulonephritis. In addition to the imaging findings, this patient's cough, dyspnea, and fever are also consistent with diffuse alveolar hemorrhage. This potentially life-threatening and organ-threatening condition has a wide array of causes, with antineutrophil cytoplasmic antibody (ANCA)–associated vasculitis being the most common.[1,2] Other, less common causes of pulmonary–renal syndrome include anti–glomerular basement membrane disease, systemic lupus erythematosus (SLE), antiphospholipid syndrome, and cryoglobulinemic vasculitis.[3]

Although pulmonary–renal syndrome is an important consideration in this patient, several aspects of her presentation are atypical. Hemoptysis may be absent in up to 30% of patients with pulmonary–renal syndrome, but the lack of hemoptysis in this patient warrants careful consideration of alternative diagnoses.[4,5] The presence of venous thromboembolism is also a potential clue since venous thromboembolism has been reported in patients with pulmonary–renal syndrome, but it is uncommon.[6,7]

Are there other possible explanations for this patient's pulmonary symptoms and the findings on imaging studies? Pulmonary embolism could cause fever and hypoxemia; however, it would not explain the findings of bilateral consolidative opacities or interlobar thickening with a ground-glass appearance. This patient reported multiple episodes of vomiting before the onset of her pulmonary symptoms, and aspiration pneumonitis or pneumonia could be an alternative explanation for the fever, cough, and pulmonary findings. I would perform testing for antimyeloperoxidase and anti–proteinase 3 antibodies to evaluate for ANCA-associated vasculitis, as well as disease-specific testing for the other potential causes of pulmonary–renal syndrome described above; however, I will now focus on diseases that cause heavy proteinuria and are associated with venous thromboembolism.

NEPHROTIC SYNDROME

Nephrotic syndrome is defined by heavy proteinuria (urinary protein excretion of >3.5 g per day), hypoalbuminemia, and peripheral edema. Although this patient did not have peripheral edema, the episodes of vomiting and recurrent nausea may have resulted in a volume-depleted state, which can mask clinically detectable edema.[8] In this patient, urinalysis showed more than 100 red cells per high-power field. Microscopic hematuria of this degree is not a typical finding in patients with nephrotic syndrome, but it could be explained by venous congestion from bilateral renal vein thrombi.[9]

Hypercoagulable State

Venous and arterial thromboemboli are well-documented complications of nephrotic syndrome.[10-12] Reported incidences of thrombotic events among patients with nephrotic syndrome vary, with deep venous thrombosis occurring in approximately 15% of patients, pulmonary embolism in 10 to 30%, and renal vein thrombosis in 25 to 37%.[13] Among patients with nephrotic syndrome, the overall incidence of thrombotic events is higher in adults (25%) than children (3%).[10] In adults, the majority of venous thromboembolic events occur within the first 6 months after the diagnosis of nephrotic syndrome.[14] A low level of serum albumin is a strong independent predictor of venous thromboembolism in patients with nephrotic syndrome. The risk of venous thromboembolism among patients with serum albumin levels below 2.5 g per deciliter is almost three times that among patients with levels above 4 g per deciliter.[15] The risk of thrombosis also varies among the underlying causes of nephrotic syndrome but is typically highest in patients with membranous nephropathy.[16,17]

The hypercoagulable state associated with nephrotic syndrome is thought to arise from a combination of factors, including genetic predisposition, localized clotting activation within the kidney, and increased number, activation, and aggregability of platelets. Hypercoagulability is primarily driven by the glomerular loss of anticoagulant and profibrinolytic proteins, alongside increased hepatic synthesis of procoagulant proteins such as fibrinogen, factor V, and factor VIII.[13]

Causes of Nephrotic Syndrome

A variety of underlying disease processes can cause nephrotic syndrome. This patient had no known diabetes, a common cause of nephrotic syndrome, and no history of a chronic inflammatory condition (e.g., chronic osteomyelitis or inflammatory bowel disease), which can lead to secondary amyloidosis and nephrotic syndrome. Class V lupus nephritis caused by SLE is an important consideration in this young patient, and associated antiphospholipid antibody syndrome could have contributed to the risk of venous thromboembolism.[18,19] However, she had no history of arthralgia, malar rash, oral sores, or cytopenias.[18] Given the absence of these clinical features, along with the absence of previous miscarriages or arterial or venous clots, SLE is unlikely to be the underlying cause of nephrotic syndrome in this patient. Therefore, I will focus on the most common causes of nephrotic syndrome — minimal change disease, focal segmental glomerulosclerosis (FSGS), and membranous nephropathy.

MINIMAL CHANGE DISEASE AND FOCAL SEGMENTAL GLOMERULOSCLEROSIS

Minimal change disease and FSGS are podocytopathies that commonly result in nephrotic syndrome. Both conditions are characterized by the histopathological finding of podocyte foot process effacement on electron microscopy.

This patient's sister had end-stage kidney disease at a young age and unfortunately died at 18 years of age. This family history suggests a possible underyling genetic cause of FSGS in this patient. However, the specific cause of her sister's kidney disease was unknown. Because this patient identifies as Hispanic, another consideration is apolipoprotein L1 (APOL1)–mediated kidney disease. Variants in the gene encoding

APOL1 have been linked to an increased risk of chronic kidney disease in persons of African ancestry, as well as in adults who identify as Hispanic or Latino.[20]

Patients with primary minimal change disease or FSGS typically present with abrupt-onset nephrotic syndrome in which peripheral edema is a predominant feature. Venous thromboembolism is rare. Given the lack of abrupt-onset peripheral edema and the rarity of venous thromboembolism in patients with minimal change disease or FSGS, these conditions are unlikely to explain this patient's presentation.

MEMBRANOUS NEPHROPATHY

One of the most common causes of nephrotic syndrome in adults, membranous nephropathy is frequently associated with venous thromboembolism.[16] Membranous nephropathy is characterized by deposition of immune complexes along the glomerular basement membrane, which compromises podocyte integrity and results in proteinuria.

Membranous nephropathy is most often a primary disorder in adults. Secondary causes include autoimmune diseases (e.g., membranous lupus nephropathy), infections (e.g., hepatitis B, hepatitis C, and syphilis), cancer (e.g., solid tumors), complications of stem-cell transplantation (e.g., graft-versus-host disease), medications (e.g., nonsteroidal antiinflammatory drugs and penicillamine), and dietary supplements (e.g., alpha lipoic acid).[21] In approximately 70 to 80% of patients, primary membranous nephropathy is caused by autoantibodies directed against the M-type phospholipase A2 receptor (PLA2R) on podocytes.[22] Other target antigens, such as thrombospondin type-1 domain-containing 7A and neural epidermal growth factor–like 1, have also been identified but are much less common causes of membranous nephropathy.[23]

A kidney biopsy remains the diagnostic standard for membranous nephropathy. However, serum testing for anti-PLA2R antibodies is sensitive and highly specific for the diagnosis of membranous nephropathy, and it allows for noninvasive diagnosis in patients with nephrotic syndrome, preserved kidney function, and no evidence of other causes, such as diabetes or cancer.[24]

The absence of other potential causes in this patient's history and initial laboratory results makes secondary membranous nephropathy an unlikely diagnosis. Therefore, nephrotic syndrome with primary membranous nephropathy and the associated complication of venous thromboembolism is the most likely explanation for this patient's presentation. Serum for anti-PLA2R antibody testing was obtained as the next diagnostic step.

DR. ANUSHYA JEYABALAN'S DIAGNOSIS

Phospholipase A2 receptor–associated membranous nephropathy.

DIAGNOSTIC TESTING

Dr. Claire Trivin-Avillach: The diagnostic tests in this case were a serum enzyme-linked immunosorbent assay (ELISA) and indirect immunofluorescence testing for circulating anti-PLA2R antibodies. Indirect immunofluorescence testing detects the presence of circulating anti-PLA2R autoantibodies with the use of cultured cells expressing recombinant PLA2R and fluorescently labeled secondary antibodies to visualize the antibody binding under a fluorescence microscope. In this patient, indirect immunofluorescence testing was positive for circulating anti-PLA2R antibodies. The ELISA was positive for anti-PLA2R antibodies, with a level of 400.3 RU per milliliter (reference range, <14).

A positive serum test for anti-PLA2R antibodies has a high specificity, reported to be 99%, for PLA2R-associated membranous nephropathy.[25] Rare cases of false positive ELISA results with an elevated level of anti-PLA2R antibodies have been reported; however, in those cases, indirect immunofluorescence testing was negative.[26,27] The sensitivity of a positive serologic test is approximately 67%; in rare cases, negative serologic tests can be seen in the early phase of disease. After immunologic remission, serologic tests will be negative despite staining for PLA2R in membranous deposits since antibodies are no longer circulating. In this patient, the positive results on both indirect immunofluorescence testing and

ELISA, taken together with the clinical context of the recent onset of nephrotic syndrome, are diagnostic for PLA2R-associated membranous nephropathy.

LABORATORY DIAGNOSIS

Phospholipase A2 receptor–associated membranous nephropathy.

DISCUSSION OF MANAGEMENT

Dr. Laurence H. Beck, Jr.: The finding of anti-PLA2R antibody seropositivity in this patient is very helpful for further disease management and highly suggestive of primary autoimmune membranous nephropathy, thereby limiting the need to aggressively search for occult cancers or other processes that might be secondarily associated with different forms of membranous nephropathy.[28] Although the ELISA for anti-PLA2R antibodies is approved by the Food and Drug Administration only for diagnosis, it is widely used off-label in the clinical setting for monitoring the course of disease and response to therapy.

Patients with nephrotic syndrome should receive supportive disease management with proteinuria-reducing agents (angiotensin-converting–enzyme inhibitors, angiotensin-receptor blockers, or sodium–glucose cotransporter 2 inhibitors), diuretics and dietary sodium restriction, and additional agents as needed to treat hypertension and hyperlipidemia. In patients with clinically evident venous thromboembolic disease, as in this case, or in those at high risk for venous thromboembolism due to severe hypoalbuminemia, heparin products or warfarin are the anticoagulant agents of choice.

The decision to start immunosuppressive therapy for membranous nephropathy has traditionally been based on urinary protein levels after 6 months of supportive care. Those with sustained proteinuria of more than 8 g per day are at high risk for progression to end-stage kidney disease, whereas those with proteinuria of 4 to 8 g per day are at moderate risk.[29] Immunosuppression is indicated for patients at high risk and may be considered for those at moderate risk. Life-threatening complications of nephrotic syndrome (such as the thromboembolic events observed in this patient) or unexplained decline in kidney function would be indications for administering treatment more urgently, without waiting 6 months. The risk categories have recently been modified for the inclusion of severe hypoalbuminemia (<2.5 g per deciliter) and, when available, anti-PLA2R antibody levels.[30] Consideration of the trajectory of anti-PLA2R antibody levels can also be used in the decision-making process; if levels of anti-PLA2R antibodies decline spontaneously, treatment with immunosuppression may not be necessary.

From a historical perspective, a 6-month immunosuppression protocol involving alternating months of glucocorticoids and alkylating agents (e.g., cyclophosphamide is used in the modified Ponticelli regimen) remains the only regimen that has been shown to preserve function at 10 years and can rapidly reduce anti-PLA2R antibody levels.[31,32] Although it is still used in patients with severe and high-risk disease, the toxic effects associated with this protocol (including gonadal suppression, which would be relevant for this young patient) have led to the development of less toxic therapies.

Calcineurin inhibitors, such as cyclosporine and tacrolimus, are used either as monotherapy or with low-dose glucocorticoids and are associated with a clinically significant risk of relapse when the agent is tapered and stopped.[33] Anti-CD20 agents, such as rituximab, are the latest and current first-line therapy. A pivotal trial showed that rituximab was superior to cyclosporine in maintaining remission of proteinuria in patients with membranous nephropathy at 24 months.[34] Although the two agents had similar efficacy at 12 months (60% and 52%, respectively), relapses that occurred in the cyclosporine group lowered the percentage of patients with a response (complete or partial remission) to 20% (with no complete remissions) at 24 months. In the rituximab group, a response occurred in 60% of patients, with complete remission in 35%.

It is important to regularly monitor for response and toxic effects with any immunosuppression regimen. A noteworthy and clinically vexing characteristic of membranous nephropathy is that the duration over which remission of proteinuria occurs is typically prolonged owing to the time needed to stop the formation of subepithelial immune complexes and the resultant complement activation, as well as the time needed for the recovery of the podocyte and glomerular filtration barrier.[35] Although complete remission at 24 months occurs in only 30 to 60% of patients, it has been used as a surrogate for longer-term clinical benefit end points and is the goal of treatment.[36] In addition to monitoring proteinuria and the level of serum albumin, some have suggested monitoring the level of anti-PLA2R antibodies as an earlier sign of treatment efficacy.[30,37]

A combination therapy regimen that includes cyclophosphamide, glucocorticoids, and rituximab, which is used at this hospital for several types of glomerular disease, has shown remarkable efficacy in an observational cohort study of membranous nephropathy, with complete remission occurring in 83% of patients at 24 months.[38] Such an aggressive regimen is not needed in all cases of membranous nephropathy; however, it would be a reasonable regimen in this case owing to the high-risk features and pulmonary embolic disease in this patient.

Dr. Jeyabalan: This patient was classified as having a high risk of disease progression owing to multiple venous thromboembolism events and a urinary protein excretion of greater than 4 g per day. For her extensive venous thromboembolism, the patient initially received treatment with low-molecular-weight heparin, which was later transitioned to warfarin. For her PLA2R-associated membranous nephropathy, she received treatment with a combination of rituximab, low-dose oral cyclophosphamide, and prednisone, along with prophylactic trimethoprim–sulfamethoxazole, omeprazole, and calcium with vitamin D supplementation. In addition, she received prescriptions for losartan and low-dose oral furosemide, and she was advised to follow a low-sodium diet. Treatment for dyslipidemia was not needed.

The patient's fever and flank pain resolved within 1 week after presentation. For aspiration pneumonia, she initially received treatment with intravenous vancomycin and cefepime, which was later transitioned to ceftriaxone and azithromycin. She was weaned off supplemental oxygen with the aid of incentive spirometry and diuretics. Laboratory test results showed an improvement in the serum creatinine level to 0.5 mg per deciliter during her hospitalization, and she was discharged home on hospital day 14.

Two months after the patient started treatment, the serum test for anti-PLA2R antibodies was negative, indicating immunologic remission. Within 3 months after starting immunosuppression and supportive measures, the urinary protein level had decreased to 1.79 g per day, indicating partial proteinuric remission. Venous thromboembolism was considered to be provoked in the context of membranous nephropathy, and treatment with warfarin was discontinued after three consecutive normal serum albumin levels (>3.5 g per deciliter). She continues to receive treatment with rituximab, which is planned for a duration of 2 years. After treatment with rituximab is stopped, the serum anti-PLA2R antibody level and urinary protein levels will be monitored every 3 to 4 months for potential relapse.

Dr. Natalia Sutherland (Medicine): Given the role of complement activation in the mechanism of disease, has inhibition of complement been used therapeutically for PLA2R-associated membranous nephropathy?

Dr. Beck: Current therapies for membranous nephropathy are largely designed to reduce circulating autoantibodies such as anti-PLA2R antibodies, but thought has been given to use of complement inhibitors that could rapidly stop complement generation and the known injurious effects of C3a, C5a, and the C5b-9 membrane attack complex on the podocyte.[39,40] Although they are still in development for this indication, complement inhibitors could potentially be a useful adjunctive therapy to lower proteinuria while immunosuppressive therapies eliminate circulating autoantibodies.

FINAL DIAGNOSIS

Phospholipase A2 receptor–associated membranous nephropathy.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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